RYERSON UNIVERSITY

DEPARTMENT OF MATHEMATICS, PHYSICS & COMPUTER SCIENCE

PCS 125 FINAL EXAMINATION

DURATION: 2.5 hours

DATE: 6 December 2002

Student Number	Section	PRINT first name	PRINT last name

Please circle your professor's name: M. Buckby W. Que R. Szmidt M. Kolios J. Schmid B. Todorow

EXAMINATION INSTRUCTIONS

1. This is a closed book, multiple-choice exam; please select the option which is closest to the correct answer. Circle correct answers in this booklet.
2. Transcribe your answers onto the bubble sheet; only the bubble sheet will be marked.
3. Do not fold, spindle or mutilate your bubble sheet.
4. The formula sheet at the end of the exam may be detached and retained. The remaining exam pages must not be unstapled. Turn bubble sheet face down when not working on it.
5. Only non-programmable calculators are permitted.
6. The exam consists of 23 questions, totalling 22 marks, and is worth 50% of your course grade.
7. The first question is for administrative purposes only. All subsequent questions are of equal weight.
8. Incorrect answers will receive a mark of zero; there are no further punitive deductions.
9. At the end of the exam, insert your bubble sheet into the exam booklet and hand in both.

Please sign here indicating you have read and understood the above instructions.

Signature: _____

(no signature, no mark!)

DO NOT OPEN THE EXAM UNTIL YOU ARE TOLD TO DO SO.

Please choose the letter of the answer which is closest to the correct result.

1) What colour is your examination paper?

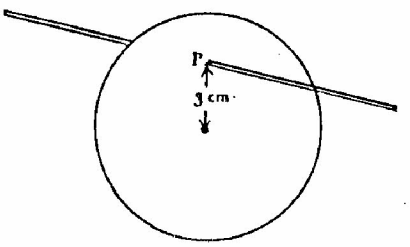
A) White B) Tan

2) A sinusoidal wave is travelling along a rope. The oscillator that generates the wave completes 60 vibrations in 30 seconds. A particular maximum (crest) travels 4.50 meters along the rope in 15 seconds. What is the wavelength (in m)?

A) 6.66 B) 2.50 C) 1.66 D) 0.60 E) 0.15

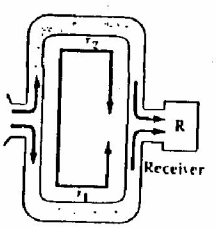
3) A solid sphere of uniform density has a moment of inertia about an axis through its centre equal to $\frac{2}{5}MR^2$. Such a sphere (R = 5.00 cm, M = 700 g) is to be used as a physical pendulum free to swing back and forth about a fixed horizontal axis which passes through point P located 3.00 cm above the centre of the sphere. Determine the period of oscillation (in seconds).

A) 0.366 B) 0.505 C) 0.531 D) 1.10 E) 5.05



4) Sound from a speaker enters a tube and follows two paths, r_1 = 1.00 m and r_2 = 0.60 m. The two sound waves superimpose at the receiver. If the speed of sound is 340 m/s, what is the lowest speaker frequency (in Hz) that will result in an intensity maximum at the receiver?

A) 85 B) 170 C) 425 D) 850 E) 1700



5) The periodic variation in sound intensity at a given point due to the superposition of two sound waves which have slightly different frequencies is called _____.

A) Resonance B) Damping C) Huygens' Principle D) The Doppler Effect

E) Beating

6) Which one of the following statements is true concerning the magnitude of the electric field at a point in space?

A) It is a measure of the total charge on the object.
B) It is a measure of the electric force on any charged object.
C) It is a measure of the ratio of the charge on an object to its mass.
D) It is a measure of the electric force per unit mass on a test charge.
E) It is a measure of the electric force per unit charge on a test charge.

7) A light ray initially in water ($n = 1.33$) enters a slab of clear plastic, which is immersed in the water, at an angle of incidence of 33.0°. Its angle of refraction in the plastic is 22.4°. Determine the speed of light (in m/s) in the plastic.

A) 2.99×10^8 B) 2.58×10^8 C) 2.18×10^8 D) 1.58×10^8 E) 1.18×10^8

8) Monochromatic light from a sodium vapor lamp ($\lambda = 589.3$ nm) is directed through a diffraction grating, producing an interference pattern on a screen which is 1.46 m from the grating. The central and first order maxima are separated by 0.45 m on the screen. Determine the number of lines per cm on the diffraction grating to within 1% accuracy.

A) 10 000 B) 7500 C) 5000 D) 2500 E) 1000

9) In approximately five billion years our Sun will exhaust its nuclear fuel and collapse to a White Dwarf. This is a dead star that has the mass of the Sun compressed into an Earth-sized volume. What is the minimum speed necessary (in m/s) for a 4000 kg space-probe to escape the Sun's surface during its White Dwarf phase?

Data: Mass of Sun = 2×10^{30} kg ; Radius of Sun = 7×10^8 m
Mass of Earth = 5.98×10^{24} kg ; Radius of Earth = 6.37×10^6 m

A) 1.80×10^5 B) 1.80×10^6 C) 4.57×10^6 D) 6.47×10^6 E) 2.99×10^8

10) What is the gravitational acceleration (in m/s²) at the surface of the White Dwarf described in Question # 9?

A) 2.1×10^{13} B) 4.6×10^6 C) 1.1×10^{13} D) 3.3×10^6 E) 3.1×10^{13}

11) A laboratory-produced charged particle interacts with an identical particle. Researchers discover that the gravitational force between the two particles has the same value as the electrostatic force between them, i.e., $F_g = F_e$. What is the charge-to-mass ratio $\left(\frac{q}{m}\right)$ for this type of particle expressed in $\frac{C}{kg}$?

A) 1.16×10^{-10} B) 8.61×10^{-11} C) 1.08×10^5 D) 1.16×10^{10}
E) 7.42×10^{-21}

12) A charged insulator is a solid sphere of R = 50.0 cm radius. A total positive charge of 25.0 μC is uniformly distributed throughout its volume. Calculate the magnitude of the electric field (in N/C) at r = 25.0 cm and at r = 75.0 cm from the center of the sphere.

A) 4.5×10^5 ; 4.0×10^5 B) 4.0×10^5 ; 4.5×10^5 C) 450 ; 400
D) 1.33×10^5 ; 8.99×10^5 E) 8.99×10^5 ; 1.33×10^5

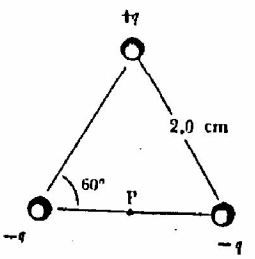
13) A $+9\mu C$ test charge is placed in a uniform electric field of 400 V/m. What is the magnitude of the electrostatic force (in N) on the test charge?

A) 4.4×10^{7} B) 2.3×10^{-11} C) 2.3×10^{-8} D) 3.6×10^{-6}

E) 3.6×10^{-3}

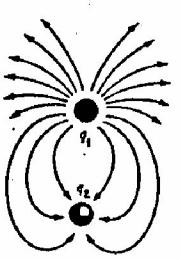
14) The three charges in the figure are at the vertices of an equilateral triangle. Each side is 2.0 cm long. If $q = 4.0\ \mu C$, calculate the absolute value of the electric field (in V/m) at the midpoint, P, of the base.

A) 2.1×10^{6} B) 1.2×10^{8} C) 7.3×10^{8} D) 1.2×10^{5} E) 7.3×10^{5}



15) The figure shows the electric field lines for two point charges separated by a small distance. Determine the absolute value of the ratio q_1 / q_2.

A) $\frac{1}{2}$ B) 2 C) $\frac{1}{3}$ D) 3 E) $\frac{1}{6}$



16) A + 0.5 nC particle is accelerated through a potential difference of 2000 V. What is the increase in its potential energy (in μJ)?

A) 1.0×10^{-3} B) 1.0×10^{3} C) 1.0×10^{-6} D) 1.0 E) 250

17) In an xy coordinate system, a uniform electric field of magnitude 300 V/m is in the positive x-direction. A 6.00 μC charge moves from rest at point A = (0.10, 0.10) m to point B = (0.20, 0.20) m. Through what potential difference (in V) did the 6.00 μC charge move?

A) -180 B) -60 C) -30 D) -10 E) -3

18) A proton enters a magnetic field which is perpendicular to its velocity vector. The proton moves in a circular path while it is in the magnetic field. If the proton requires 2.00 μs to complete one circular revolution, find the magnitude of this magnetic field (in T).

A) 1.8×10^{-5} B) 3.3×10^{-2} C) 1.8×10^{-2} D) 33 E) 18

19) In a right-handed xyz coordinate system, a proton moves in the positive x-direction through a region of uniform electric field, $\vec{E}$, which is in the negative y-direction and magnetic field, $\vec{B}$. The proton experiences a constant acceleration in the positive y-direction. What is the direction of the electrostatic force, $\vec{F}_e$, experienced by the proton? What is the direction of the magnetic field, $\vec{B}$?

A) y ; x B) y ; z C) z ; y D) -y ; -z E) -z ; -y

20) A velocity selector has a magnetic field of 1.00×10^{-2} T. If 10.0 keV electrons are to pass through undeflected, what electric field strength (in V/m) is required?

A) 5.93×10^{7} B) 5.93×10^{5} C) 4.19×10^{5} D) 1.38×10^{4} E) 1.48×10^{15}

21) A positively charged particle carries 20 e (where e is the fundamental charge) and moves with a velocity of $\vec{v} = (2\vec{i} - 4\vec{j} + \vec{k})$ m / s in a region where the field is $\vec{B} = (\vec{i} + 2\vec{j} - 3\vec{k})$T. What is the magnitude of the magnetic force (in N) experienced by this charge?

A) 2.34×10^{-18} B) 6.40×10^{-18} C) 2.81×10^{-18} D) 4.67×10^{-17}

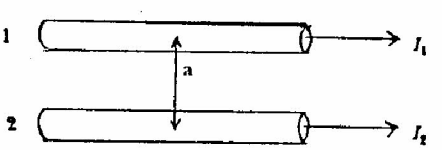
E) 2.34×10^{-15}

22) A long, thin, straight conducting wire carries a current of 10.0 A. What distance (in cm) from the central axis of the wire is the magnetic field 1.00×10^{-4} T?

A) 1.00 B) 0.01 C) 0.20 D) 0.02 E) 2.00

23) Two parallel wires, each carrying a steady current, I, are separated by a center-to-center distance, a. The magnetic field $\vec{B}_2$ arising from the current in wire 2 exerts a force on wire 1 which has a magnitude, $F_1 = I_1 L B_2$ (and vice versa). If $I_1 = 2A$ and $I_2 = 6A$, which statement is true?

A) $F_1 = 3F_2$ B) $F_1 = \frac{F_2}{3}$ C) $F_1 = F_2$ D) $F_1 = 3aF_2$ E) $F_1 = \frac{a}{3}F_2$



CONSTANTS AND FORMULAE

$G = 6.672x10^{-11} N \cdot m^2/kg^2$

$g = 9.80 m/s^2$

$e = 1.60x10^{-19} J$

$1eV = 1.60x10^{-19} J$

$k = 8.99x10^9 N \cdot m^2 / C^2 = \frac{1}{4\pi \epsilon_0}$

$\epsilon_0 = 8.85x10^{-12} C^2/(N \cdot m^2)$

$\mu_0 = 4\pi x10^{-7} N/A^2 = 1.26x10^{-6} N/A^2$

$v_{sound} = 340 m/s$

$k' = \mu_0/4\pi = 10^{-7} N/A^2$

$1T = 10^4 G$

$c = 3.00x10^8 m/s$

$R_H = 1.097x10^7/m$

$h = 6.63x10^{-34} J \cdot s$

$h = 4.14x10^{-15} eV \cdot s$

$R_E = 6.37x10^6 m$

$M_E = 5.98x10^{24} kg$

$R_{moon} = 1.74x10^6 m$

$M_{moon} = 7.36x10^{22} kg$

$D_{E-moon} = 3.84x10^8 m$

$m_p = 1.67x10^{-27} kg$

$m_e = 9.11x10^{-31} kg$

$$\cos a + \cos b = 2\cos\frac{a+b}{2}\cos\frac{a-b}{2} \qquad \sin a + \sin b = 2\sin\frac{a+b}{2}\cos\frac{a-b}{2}$$

$\omega = \omega_0 + \alpha t$

$\theta - \theta_0 = \omega_0 t + \frac{1}{2}\alpha t^2 = \frac{1}{2}(\omega_0 + \omega)t$

$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$

$x = A\cos(\omega t + \phi)$

$v = \frac{dx}{dt} = -\omega A \sin(\omega t + \phi)$

$a = \frac{dv}{dt} = \omega^2 A \cos(\omega t + \phi) = -\omega^2 x$

$U = \frac{1}{2}kx^2 \quad E = \frac{1}{2}kA^2$

$\omega = \sqrt{\frac{k}{m}} \quad \omega = \sqrt{\frac{g}{L}} \quad \omega = \sqrt{\frac{mgd}{I}}$

$T = \frac{2\pi}{\omega} \quad T = 2\pi\sqrt{\frac{m}{k}} \quad T = 2\pi\sqrt{\frac{L}{g}} \quad T = 2\pi\sqrt{\frac{I}{mgd}}$

$v = \lambda f \quad v = \sqrt{\frac{F}{\mu}} \quad v = \frac{\lambda}{T}$

$k = \frac{2\pi}{\lambda}$

$\frac{d^2\theta}{dt^2} = \frac{-g\theta}{L}$

$\theta = \theta_{max}\cos(\omega t + \phi)$

$y(x,t) = A\sin(kx \pm \omega t + \phi)$

$Power = \frac{1}{2}\mu\omega^2 A^2 v$

$y = 2A_0 \cos\left(\frac{\phi}{2}\right)\sin\left(kx - \omega t - \frac{\phi}{2}\right)$

$y = (2A_0 \sin kx)\cos\omega t$

$f_n = \frac{v}{\lambda_n} = \frac{n}{2L}v$

$v = \sqrt{\frac{B}{\rho}}$

$v_D = c\frac{\Delta\lambda}{\lambda}$

$f' = f\left(\frac{v \pm v_0}{v \mp v_s}\right)$

$\vec{S} = \frac{1}{\mu}\vec{E}x\vec{B}$

$E = hf \quad n = \frac{c}{v} \quad \lambda_n = \frac{\lambda_0}{n}$

$\lambda_1 n_1 = \lambda_2 n_2$

$n_1 \sin\theta_1 = n_2 \sin\theta_2$

$n = \frac{\sin\left(\frac{\theta + \delta_{min}}{2}\right)}{\sin(\theta/2)}$

$\sin\theta_c = n_2/n_1 \quad (n_1 \geq n_2)$

$\delta_c = d\sin\theta = m\lambda$

$\delta_d = d\sin\theta = \left(m + \frac{1}{2}\right)\lambda$

$2nt = \left(m + \frac{1}{2}\right)\lambda$

$2nt = m\lambda$

$R = Nm = \frac{\lambda}{\Delta\lambda}$